Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – XI

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Class: TYIT	Batch:01
Date of Assignment: 01-09-2026	Date/Time of Submission:01-09-2026

Practical 11: Implementing Advanced Visualization Techniques and Customizations

A) Create slope charts and lollipop charts.

Steps:-

1.Open Tableau and connect to the required dataset, then create a new worksheet.

2.For a **Slope Chart**, drag the Category field to **Rows** and the Year/Time field to **Columns**.

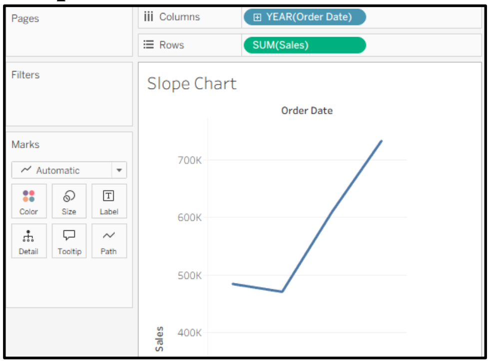
3.Drag the required Measure to the view and select **Line** from the Marks card.

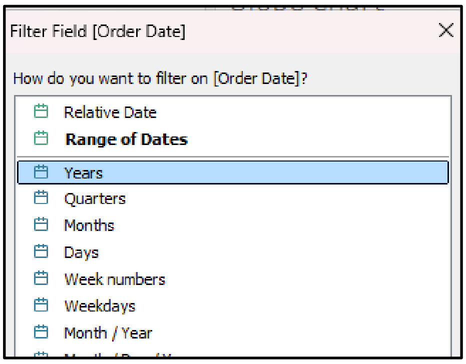
4.For a **Lollipop Chart**, place the Category on **Rows** and Measure on **Columns**.

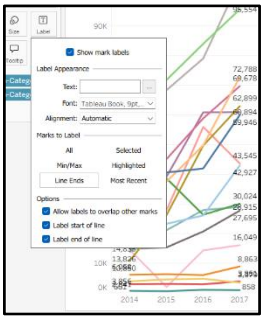
5.Create a **Dual Axis**, keep one mark as **Bar** and the other as **Circle**, and make the bar thin.

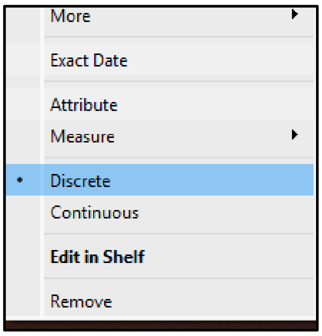
6.Add data labels and format the chart to clearly display the values.

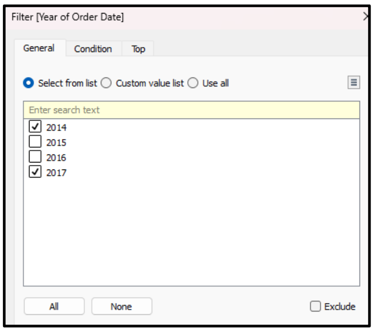
Output:-

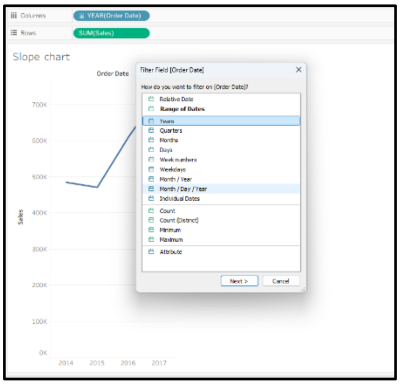


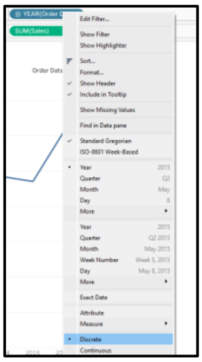


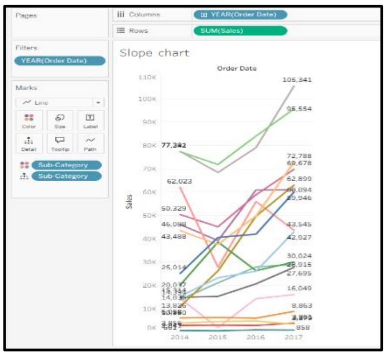




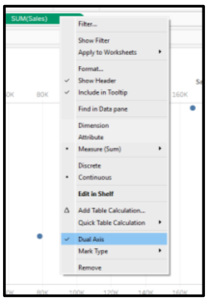
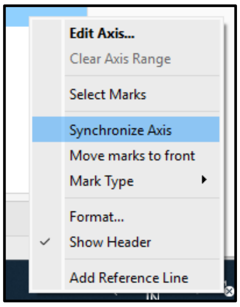


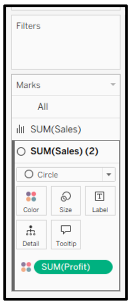


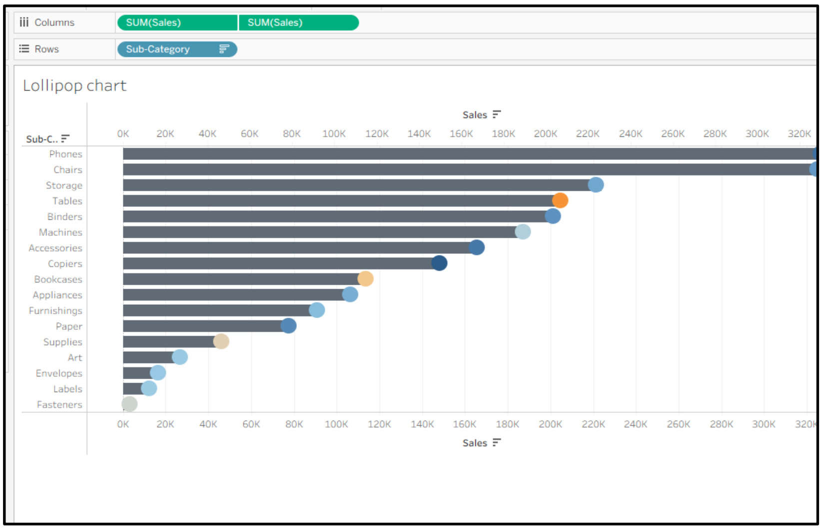




lollipop charts





B) Design waterfall charts.

Steps:-

1.Open a new worksheet and connect the required dataset.

2.Drag the Category field to Columns and the Measure field to Rows.

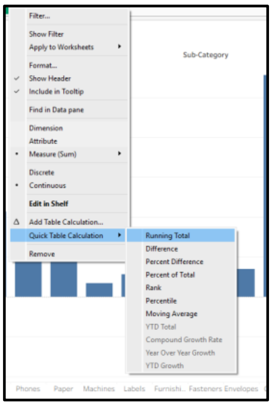
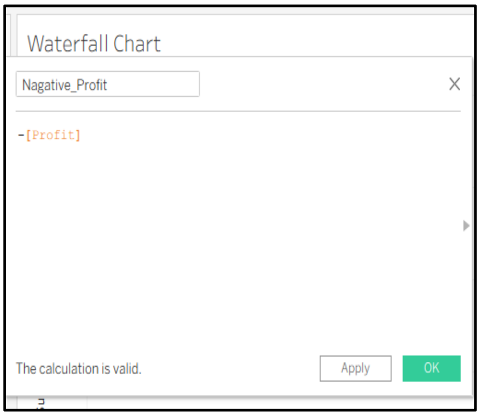
3.Create the required calculated fields to determine the starting position of each bar.

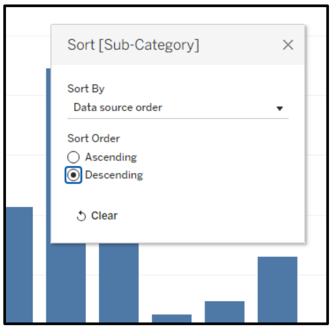
4.Change the Marks type to Gantt Bar to create the waterfall structure.

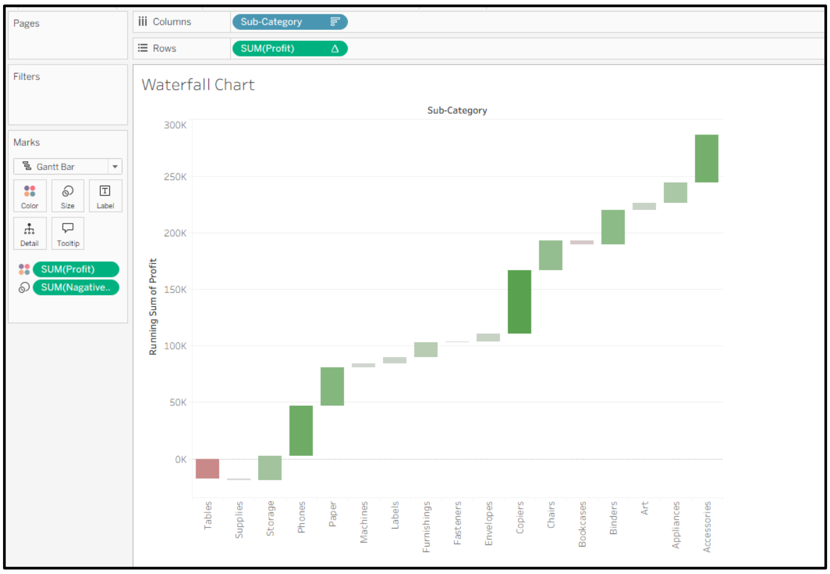
5.Apply different colors to positive and negative values for better understanding.

6.Add labels and formatting so the final chart clearly shows the overall increase, decrease, and total.

Output:-





C) Build sparklines for trend analysis.

Steps:-

1.Create a new worksheet and place the required Category field on **Rows**.

2.Drag the Date/Year field to **Columns** to display the time period.

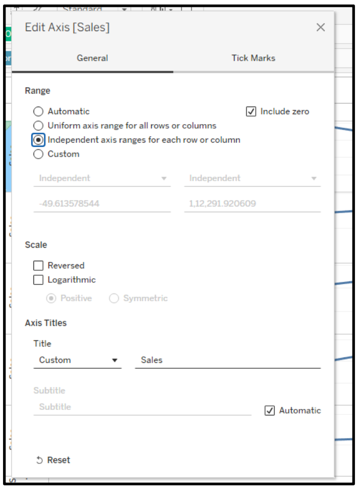
3.Drag the required Measure into the view and select **Line** from the Marks card.

4.Reduce the chart size to create a small trend-line or sparkline for each category.

5.Hide unnecessary axes, gridlines, and headers to give it a compact appearance.

6.Add the required labels or latest values and format the sparklines for easy comparison.

Output:-



D) Implement sheet swapping for dynamic dashboards.

Steps:-

1.Create two or more worksheets containing the visualizations that need to be switched.

2.Create a **Parameter** with values representing each worksheet or visualization.

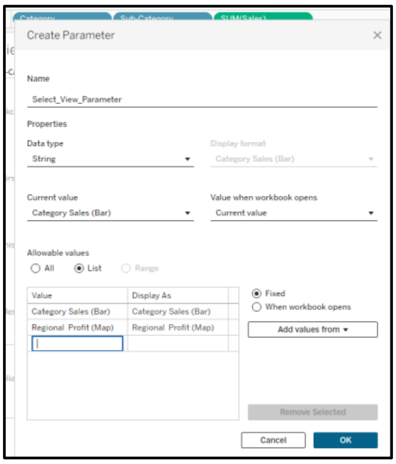
3.Create a calculated field that checks the selected parameter value.

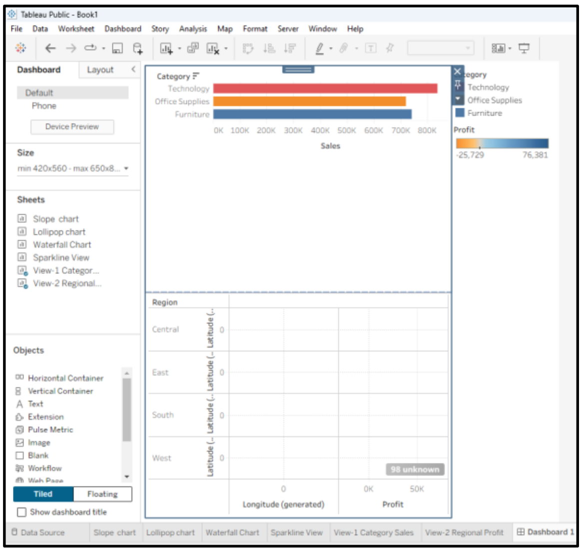
4.Add the calculated field as a filter to the respective worksheets.

5.Place the required worksheets and the Parameter control on a **Dashboard**.

6.Select different parameter values to test whether the dashboard dynamically switches between sheets.

Output:-





E) Create custom geographic territories.

Steps:-

1.Open the dataset in Tableau and identify the geographic field such as Country, State, or City.

2.Right-click the geographic field and select **Create → Group**.

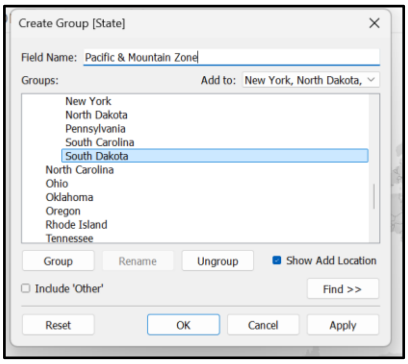
3.Select the required locations and group them into custom territories.

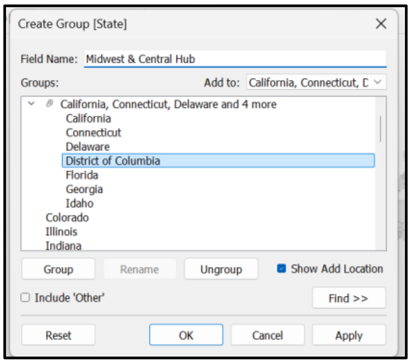
4.Rename each group according to the required territory name.

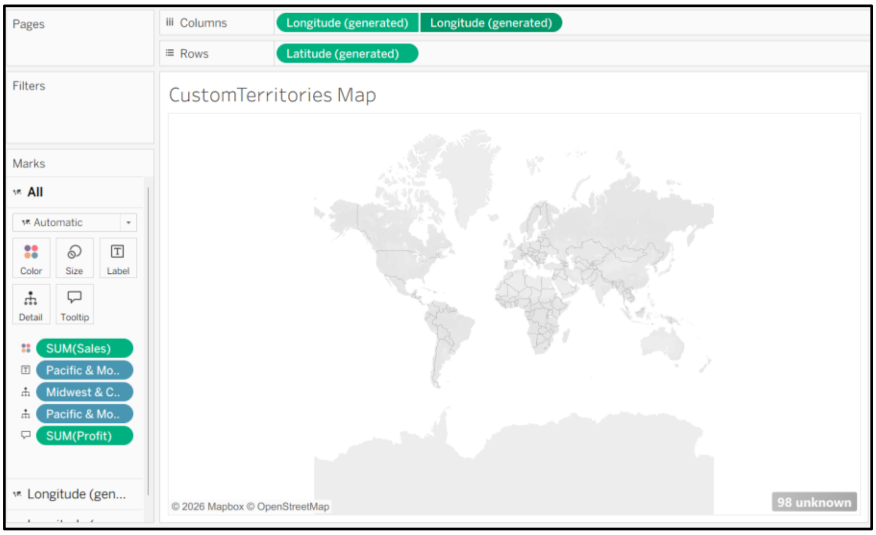
5.Drag the newly created territory field into the visualization and select the **Map** view.

6.Apply different colors to the territories and format the map for clear presentation.

Output:-







F) Use background images in dashboards.

Steps:-

1.Create a new Tableau Dashboard and prepare the required worksheets.

2.From the **Dashboard** pane, add an **Image** object.

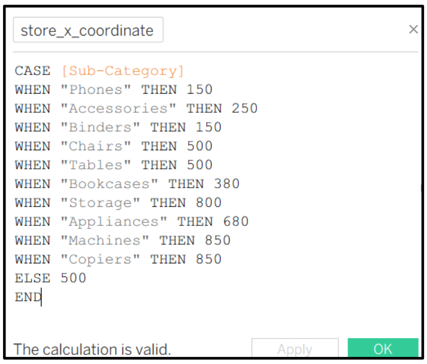
3.Select the required background image from your computer.

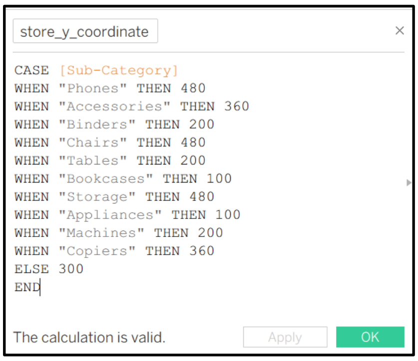
4.Resize and position the image so that it covers the required dashboard area.

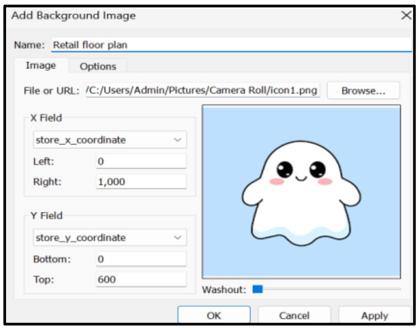
5.Place the charts and other dashboard objects over the background image.

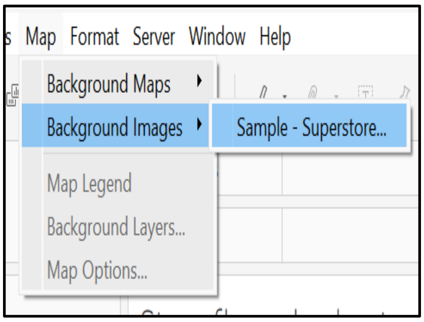
6.Adjust the layout and formatting to ensure that all charts and text are clearly visible.

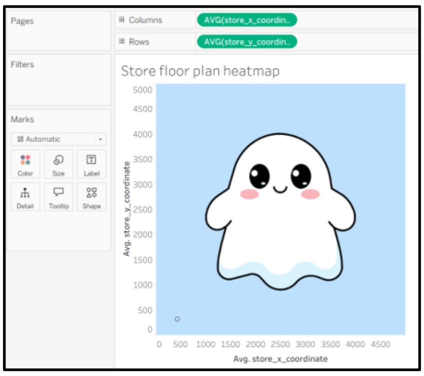
Output:-

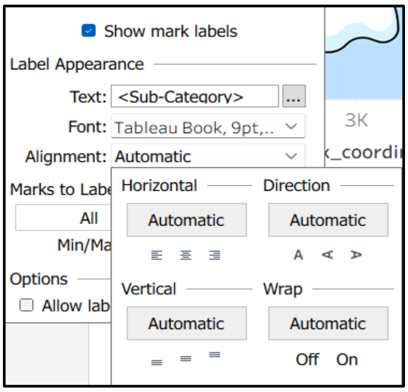












g) Apply animation and transparency techniques.

Steps:-

1.Open the required Tableau dashboard containing your visualizations.

2.Enable **Animations** from the appropriate Tableau formatting options, if supported by your version.

3.Set the animation duration and test the transition by changing filters or parameters.

4.Select the required dashboard object and open its formatting options.

5.Adjust the **shading/background transparency** to create a transparent or semi-transparent effect.

6.Test the dashboard and ensure that the animation and transparency improve the visual appearance without affecting readability.

Output:-